Higher Institute of Technological Studies of Bizerte

RAIA2 (M2-S3) — Dept. of EE | **Academic Year: 2026–2027**

Midterm Exam | DL and RL | **Date: October 29, 2026**

Teacher: A. Mhamdi | **Duration: 60 min**

This document consists of **10** pages. Please check that your copy is complete and report any missing page immediately. The **3** tasks are independent and may be solved in any order. The following rules apply:

① **One handwritten, double-sided A4 sheet** is allowed.
② **No electronic devices** may be used, except for a non-programmable calculator.
③ **Round all numerical results** to three decimal places.
④ **Justify your reasoning** to receive full credit.
⑤ **Correct solutions** earn the points indicated, with no negative marking.
⑥ **Write only** on the answer sheets: anything written on this exam paper will be ignored.

SOLUTIONS

Task №1 — Latent Semantic Analysis

⏳20 min | (4 points)

Corpus

d1	the team won the match
d2	the player scored a goal in the match
d3	the coach praised the team and the player
d4	the chef cooked pasta with tomato sauce
d5	add tomato and garlic to the sauce
d6	the chef praised the pasta and the sauce

Code

Julia

```julia
using LinearAlgebra
tokenize(s) = split(lowercase(s))
docs = [d1, d2, d3, d4, d5, d6]
vocab = sort(unique(vcat(tokenize.(docs)...)))  # 21-element Vector
A  = [count(==(w), tokenize(d)) for w in vocab, d in docs] # terms×docs
N  = size(A, 2)
df = vec(sum(A .> 0; dims=2))
X  = A .* log.(N ./ df)  # tf-idf
F  = svd(X)  # U, S, V
k  = 2
Uk, Sk, Vk = F.U[:, 1:k], F.S[1:k], F.V[:, 1:k]
```

Output (rounded to 2 decimals)

F.S

σ_1	3.95
σ_2	3.71
σ_3	3.14
σ_4	2.82
σ_5	2.27
σ_6	1.67

Vk (rows = docs)

	dim 1	dim 2
d1	0.14	0.0
d2	0.97	0.09
d3	0.17	−0.11
d4	0.04	−0.63
d5	0.05	−0.71
d6	0.04	−0.29

Uk (selected terms)

	dim 1	dim 2
match	0.31	0.03
player	0.32	−0.01
goal	0.44	0.05
chef	0.02	−0.27
pasta	0.02	−0.27
tomato	0.02	−0.4

$$\text{tf-idf}(t,d) = \text{tf}(t,d) \cdot \log\left(\frac{N}{\text{df}(t)}\right) \tag{1}$$

- $\text{tf}(t,d)$: number of times term t appears in document d *(raw count)*.[1]
- N: total number of documents in the corpus.
- $\text{df}(t)$: number of documents containing t.
- $\log\left(\frac{N}{\text{df}(t)}\right)$ *(idf)*: rewards rare terms; equals 0 when t appears in every document.

(a) (0.5 pt) Give the dimensions of A. What does `df` contain?

- A is 21 × 6 (21 terms, 6 docs);
- `df` = number of docs containing each term.

(b) (0.5 pt) What is `log.(N ./ df)` for "the"? Why?

"the" appears in all 6 docs, so df = 6 and $\log\left(\frac{6}{6}\right) = 0$. A term in every doc carries no discriminative information, so its tf-idf is 0.

(c) (0.5 pt) Compute the tf-idf of "tomato" in d5.

tf = 1, df = 2 (d4, d5), so tf-idf = $1 \cdot \log\left(\frac{6}{2}\right) = \log(3) \approx 1.10$.

(d) (1 pt) Compute the energy captured by the first two components:

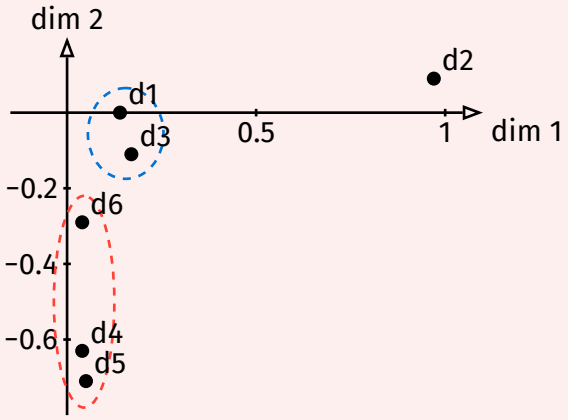
[1] Both definitions are valid. Here $\text{tf}(t,d)$ is the raw count of t in d. The normalized variant divides it by the document length: $\frac{\text{tf}(t,d)}{\sum_{t'} \text{tf}(t',d)}$.

$$\mathfrak{E}_{1,2} = \frac{\sigma_1^2 + \sigma_2^2}{\sum_j \sigma_j^2}. \tag{2}$$

Is $k = 2$ a good choice? Justify.

$(3.95^2 + 3.71^2)/55.1 \approx 53\%$. No clear elbow, since the spectrum is almost flat; an 80% threshold would need $k = 4$. The documents are short and share few words, so $k = 2$ captures the topics only partly.

(e) (0.5 pt) Using Vk, sketch the six documents in 2D and group them.



The sports docs d1 and d3 are close together near the origin, d2 is an outlier on the right, and the cooking docs form a group along the negative dim 2 axis, with d6 closest to the origin because it shares “praised” with d3.

(f) (0.5 pt) On another machine, all signs in column 1 of Uk and Vk are flipped. Does the conclusion change? Why?

No. $(u_1, v_1) \to (-u_1, -v_1)$ leaves $\sigma_1 u_1 v_1^T$, and hence X, unchanged, so signs are arbitrary. The points are mirrored left-right, which preserves distances and angles, so clusters and cosines are unchanged. Top terms by |value| are unchanged too.

(g) (0.5 pt) Using Uk, give the top 3 terms (by |value|) of each dimension among those listed, and name each topic.

Dim 1 goal (0.44), player (0.32), match (0.31): **sports**.
Dim 2 tomato (-0.40), chef and pasta (-0.27 each): **cooking**. The sign only gives the direction: cooking terms are negative on dim 2, which matches d_4, d_5, d_6 in Vk.

Task №2 — CNN and cross-entropy

⏳20 min | (6 points)

Input: 30 × 30 × 1 grayscale image, 3 classes.

#	Layer
1	Conv2D, 8 filters, 3 × 3, stride 1, no padding, ReLU
2	MaxPool 2 × 2, stride 2
3	Conv2D, 16 filters, 3 × 3, stride 1, no padding, ReLU
4	MaxPool 2 × 2, stride 2
5	Flatten
6	Dense, 3 units, no activation (logits)

(a) (0.25 pt) Give the output shape after each layer.

Output shapes 28 × 28 × 8, 14 × 14 × 8, 12 × 12 × 16, 6 × 6 × 16, 576, 3.

(b) (0.25 pt) Give the number of trainable parameters of each layer *(biases included)*.

Parameters 80; 0; 1168; 0; 0; $576 \cdot 3 + 3 = 1731$.

(c) (0.25 pt) What is the total number of parameters?

Total $80 + 1168 + 1731 = 2979$

(d) (0.25 pt) What is the output shape for a batch of 32 images?

Output shape $(32, 3)$

✳ For one image: logits $\boldsymbol{z} = [2.0, 1.0, 0.1]$, true class $k = 2$, so $\boldsymbol{y} = [0, 1, 0]$.

(e) (0.5 pt) Compute the softmax probabilities $\boldsymbol{p}$.

$$\sum_j e^{z_j} = e^2 + e^1 + e^{0.1} \approx 7.389 + 2.718 + 1.105 = 11.212 \tag{3}$$

so

$$\boldsymbol{p} \approx [0.659, 0.242, 0.099] \tag{4}$$

(f) (0.5 pt) Compute the categorical cross-entropy.

$$\begin{aligned}\mathcal{L} &= -\sum_i y_i \log(p_i) \\ &= -\log(p_1) = -\log(0.242) \approx 1.417\end{aligned} \tag{5}$$

(g) (0.5 pt) Compute the loss from logits.

$$\begin{aligned}\mathcal{L} &= -z_k + \log \sum_j \exp(z_j) \\ &= -1.0 + \log(11.212) \approx -1.0 + 2.417 = 1.417\end{aligned} \tag{6}$$

(h) (0.5 pt) Do your results to (f) and (g) match? Why?

Yes. Both are the same quantity:

$$-\log \operatorname{softmax}(z)_k = -z_k + \log \sum_j \exp(z_j) \tag{7}$$

(i) (1 pt) Compute $\frac{\partial \mathcal{L}}{\partial \boldsymbol{z}}$.

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial \boldsymbol{z}} &= \frac{\partial}{\partial \boldsymbol{z}} \sum_k \left(-y_k \times \left[z_k - \log\left(\sum_j \exp(z_j) \right) \right] \right) \\ &= \frac{\partial}{\partial \boldsymbol{z}} \left[\left(\sum_k -y_k \times z_k \right) + \log\left(\sum_j \exp(z_j) \right) \right]\end{aligned} \tag{8}$$

Compute the derivative for each k:

$$\frac{\partial \mathcal{L}}{\partial z_k} = -y_k + \underbrace{\frac{\exp(z_k)}{\sum_j \exp(z_j)}}_{p_k} \tag{9}$$

The end result is a vector of the same shape as $\boldsymbol{z}$ and is given by

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{z}} = \boldsymbol{p} - \boldsymbol{y} \approx [0.659, -0.758, 0.099] \tag{10}$$

✳ Now $\boldsymbol{z} = [1000, 999, 998]$, true class $k = 1$.

(j) (1 pt) What goes wrong when computing $\exp\left(z_j\right)$ directly?

e^{1000} overflows to ∞. The softmax then becomes $\frac{\infty}{\infty}$ = NaN.

(k) (0.5 pt) Subtract $\max(\boldsymbol{z})$ and apply the log-sum-exp trick. Compute the loss.

$$\boldsymbol{z} - \max(\boldsymbol{z}) = [0, -1, -2] \tag{11}$$

so

$$\mathcal{L} = -0 + \log\left(1 + e^{-1} + e^{-2}\right) \approx \log(1.503) \approx 0.408 \tag{12}$$

✳ A second image has logits $[0.5, 0.2, 3.0]$ and true class $k = 3$.

(l) (0.5 pt) Compute its loss, then the mean loss over both images *(the one from question (g) and this one)*.

$$\sum_j e^{z_j} = 1.649 + 1.221 + 20.086 = 22.956 \tag{13}$$

so

$$\mathcal{L}_2 = -3.0 + \log(22.956) \approx 0.134 \tag{14}$$

Mean:

$$\frac{1.417 + 0.134}{2} \approx 0.775 \tag{15}$$

❦

RAIA2 (M2-S3) — Dept. of EE	Full Name: ----------
MIDTERM EXAM	DL AND RL	ID: ----------
Teacher: A. Mhamdi	Class: ----------
October 29, 2026	Room: ----------

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> ⚠️ **TO BE HANDED IN**
>
> The pages from this point on **must be handed in** with your answer sheets, even if left blank.

TASK №3

⏳20 min | (10 points)

(a) (0.5 pt) Which CRISP-DM phase comes immediately after Data Preparation?

○ Data Understanding ✓ Modeling ○ Evaluation ○ Deployment

(b) (0.5 pt) In CRISP-DM, which phase checks whether the model actually meets the business objectives?

✓ Evaluation ○ Data Preparation ○ Modeling ○ Deployment

(c) (0.5 pt) Which TDSP lifecycle stage has no direct equivalent in CRISP-DM?

○ Business Understanding

○ Modeling

○ Deployment

✓ Customer Acceptance

(d) (0.5 pt) Which NLP task assigns a grammatical category such as noun or verb to each word?

✓ POS ○ NER ○ Sentiment analysis ○ Machine translation

(e) (0.5 pt) In `TextAnalysis.jl`, which type is used to represent a single raw text document stored as a string?

○ `CorpusDocument` ○ `TokenArray` ○ `NGramDict` ✓ `StringDocument`

(f) (0.5 pt) Which of these is a well-known CNN architecture for image classification?

○ `BERT` ✓ `ResNet` ○ `Word2Vec` ○ `GPT`

(g) (0.5 pt) Which dataset is commonly used to pretrain CNNs for transfer learning in computer vision?

SOLUTIONS

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○ MNIST ○ IMDB Reviews ○ Iris ✓ ImageNet

(h) (0.5 pt) Which architecture popularized residual connections and won ImageNet in 2015?

✓ ResNet ○ AlexNet ○ LeNet ○ VGG

(i) (0.5 pt) What is the main purpose of a pooling layer in a CNN?

○ Increase the number of channels

✓ Reduce spatial dimensions

○ Apply the activation function

○ Compute the loss

(j) (0.5 pt) Batch normalization normalizes activations across which dimension?

○ Features of a single sample

○ Layers of the network

✓ The batch, per feature/channel

○ Time steps only

(k) (0.5 pt) In fine-tuning, what is typically done with the pretrained model's weights?

○ They are discarded

✓ They are used as initialization and updated on the new task

○ They are randomized

○ They are converted to integers

(l) (0.5 pt) Which situation benefits most from transfer learning?

○ Huge labeled dataset and unlimited compute

○ Task unrelated to any pretrained model

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- ○ No data at all
- ✓ Small labeled dataset for the target task

(m) (0.5 pt) In a residual block, what is the output computed as?

○ x only ✓ $F(x) + x$ ○ $F(x) - x$ ○ $F(x) \times x$

(n) (0.5 pt) When using a pretrained CNN as a feature extractor, which part is usually replaced and trained on the new task?

- ○ The first convolutional layer
- ○ All pooling layers
- ✓ The final classification layer
- ○ The input images

(o) (0.5 pt) What problem do residual (skip) connections primarily help address in very deep networks?

- ○ Overfitting on small datasets
- ✓ Vanishing gradients and degradation
- ○ Slow data loading
- ○ Class imbalance

(p) (0.5 pt) What happens when the following code runs?

```julia
struct Box v::Vector{Int} end
b = Box([1, 2, 3])
b.v[1] = 10
b.v = [4, 5]
```

- ○ Both assignments succeed
- ✓ The first succeeds, the second throws an error

SOLUTIONS

DO NOT WRITE ANYTHING HERE

✂ ----------

- ○ The first throws an error, the second succeeds
- ○ Both assignments throw an error

(q) (0.5 pt) What do the last two lines evaluate to, in order?

```julia
struct P x::Int end
mutable struct M x::Int end
P(1) === P(1)
M(1) === M(1)
```

○ `true, true` ✓ `true, false` ○ `false, true` ○ `false, false`

(r) (0.5 pt) What happens on the last line?

```julia
mutable struct Node
    val::Int
    next::Node
    Node(v) = new(v)
end
n = Node(1)
n.next
```

- ○ Returns `nothing`
- ○ Returns `Node(0)`
- ✓ Throws `UndefRefError`
- ○ Throws `MethodError`

(s) (0.5 pt) Which file records the exact versions of all dependencies for reproducibility?

✓ `Manifest.toml` ○ `Project.toml` ○ `README.md` ○ `startup.jl`

(t) (0.5 pt) Which package is used to run Julia code in Jupyter notebooks?

○ `Pluto.jl` ✓ `IJulia.jl` ○ `Revise.jl` ○ `BenchmarkTools.jl`

SOLUTIONS